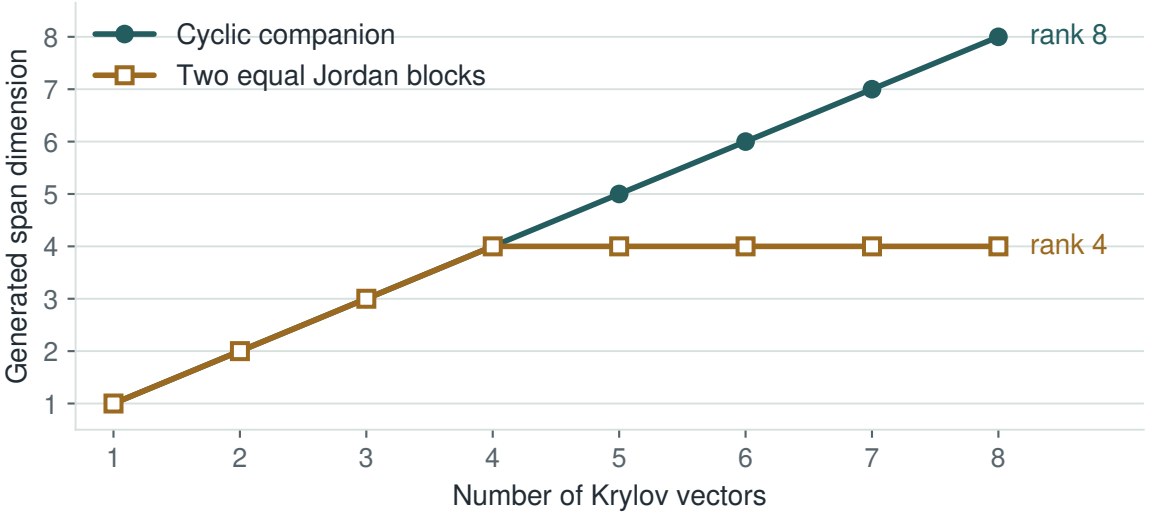
One seed can span the space or stall

Two operators on an eight dimensional space give different Krylov profiles.



The vectors are $v, Tv, \ldots, T^{r-1}v$. Both profiles use one fixed seed.